

AzuraCast Custom

Next Phase Roadmap

Clock Wheel · Station ID · Analytics · Advertising

June 2026 · v2 Complete Edition

This document is the complete record of every improvement discussed: current PR status, honest scores vs enterprise internet radio platforms, clock wheel gaps, station ID top-of-hour fix, realistic advertising options, and the full analytics build plan. Nothing from our conversation has been omitted except podcast hybrid scheduling (out of scope) and FCC/FM requirements (irrelevant to internet radio).

1 · Current PR Status

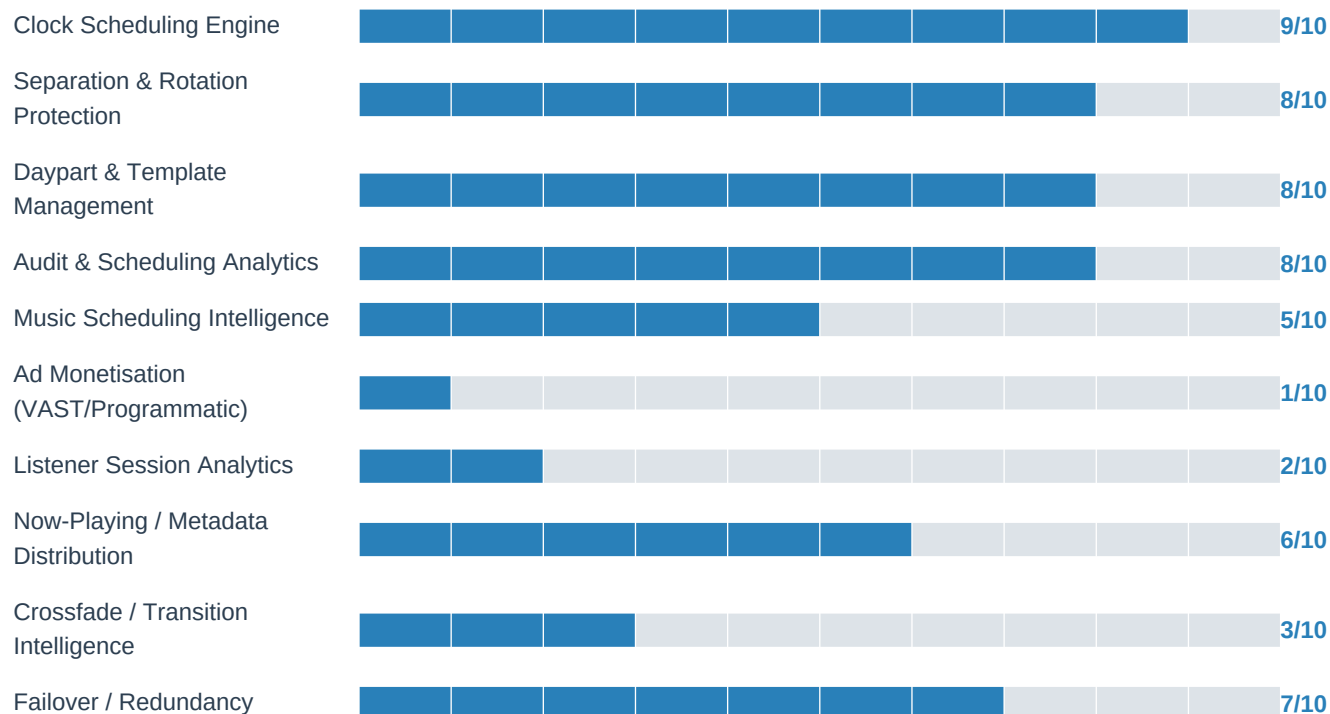
PR	Name	Status	Notes
PR 1	Timed Slot Infrastructure	■ Done	position_seconds, anchor types, slot_order all implemented
PR 2	Schedule Conflict Detection	■ Done	ScheduleConflictChecker fully implemented including emergency bypass
PR 3	Timeline Playback Engine	■ Done	ClockWheelPlaybackPlanner — fit-to-window, fallback, hard/soft anchors
PR 4	Validation & Operational Logging	■ Done	ClockWheelEventLogger, ClockWheelAnnotator (cue_out caps) wired in
PR 5	Frontend Slot Editor	■ Done	ClockWheels.vue, EditModal, Entries.vue, DaypartEditModal all present
PR 6	Automated Testing	■■ Partial	Framework wiring exists — tests/Unit folder and test classes not written
PR 7	Live Clock Wheel Tab	■ Done	ClockWheelLiveTab.vue + useClockWheelLiveData.ts complete
PR 8	Liquidsoap Alignment	■ Done	Complete per spec
PR 9	Separation Rules + Burn Rate	■ Done	SeparationRulesChecker, ClockWheelSeparationSettings, SeparationFilterResult

PR 10	Daypart Clock Inheritance	■ Done	Templates, Dayparts, InheritanceService with propagation + auto-generation
PR 11	Audit Event Database	■ Done	clock_wheel_events table, repository, logger all present
PR 12	Preview Simulator + Analytics	■ Done	PreviewAction, AnalyticsAction, ClockWheelPreviewSimulator, fill_strategy, frontend modals
PR 13	Emergency Override	■ Done	is_emergency on StationSchedule, migration, scheduler guard, frontend checkbox

12 of 13 PRs complete. PR6 is the only remaining gap — the tests/Unit folder at project root does not exist and no unit test classes have been written. The Codeception framework plumbing in backend/src/Tests/ is ready and waiting.

2 · Honest Scores vs Enterprise Internet Radio

Scored against what real commercial internet radio platforms — Triton, Shoutcast, AdsWizz-powered stations — actually deliver. Non-commercial internet radio scores higher because traffic logs, ad infrastructure, and audience reporting are optional.



Overall — Non-Commercial Internet Radio

8.5 / 10

Overall — Commercial Internet Radio (with ads + audience proof)

6.0 / 10

What You Can Drop Entirely for Internet Radio

- FCC traffic logs and commercial ad scheduling — no FCC obligations on internet radio
- Legal ID compliance logging for broadcast regulators — not required
- RDS/RBDS metadata push to FM transmitter — FM only, irrelevant
- ISDN/codec remote broadcast hardware — you are already IP native
- Hot-standby hardware failover — AzuraCast handles this reasonably already

3 · PR6 — Completing Automated Testing

What Is Missing

- The tests/Unit folder at project root does not exist
- No PHPUnit or Codeception test classes have been written for the clock wheel system
- Only a bash smoke script (util/test_clock_wheels.sh) exists — not a substitute for unit tests
- Framework plumbing in backend/src/Tests/ (Connector.php, Module.php) is ready

Test Classes Required

Test Class	What It Must Cover
ScheduleConflictCheckerTest	Same-day overlap, partial overlap, overnight overlap, recurring conflicts, emergency bypass
ClockWheelSchedulerTest	Playlists override wheels, wheels run in open windows, AutoDJ fallback fires correctly
HourTimelineTest / PlannerTest	Timeline progression, slot resolution, fit-to-window behaviour, fallback selection
SeparationRulesCheckerTest	Artist/title separation windows, relaxation cascade order, burn rate deprioritisation
ClockWheelInheritanceTest	Template propagation, instance override retention, daypart auto-generates correct hours
API Integration Tests	Slot save/reload, validation responses, conflict API, preview endpoint accuracy

4 · Clock Wheel — Remaining Gaps to Professional Standard

■ High Priority

Format Clock Category Enforcement

- Music slots currently draw from any music — a real clock wheel enforces 'slot 3 draws from Power Current only, slot 7 from Gold only'
- Slot entity has playlist targeting but it needs to be a hard constraint, not a soft preference
- Without this a PD cannot program a proper format — the single most important remaining gap in the clock wheel

Hard Anchor Miss Recovery

- When a hard anchor is missed the system currently moves on silently
- Required: log to clock_wheel_events, flag in audit table, trigger emergency AutoDJ fallback response
- PR12 already tracks missed hard anchors — add the active recovery behaviour to match

Per-Slot Separation Overrides

- Separation rules are currently wheel-level only
- Real PDs need 'this Gold slot has 7-day title separation, this Power Current slot needs only 3 hours'
- Per-slot separation config is standard in MusicMaster and Selector

■ Medium Priority

Rotation Goals — Positive Scheduling

- Burn rate protection stops over-playing (ceiling) — already built in PR9
- Missing: positive rotation goals — 'every song in Power Current must play at least once every 24 hours'
- Both ceiling and floor are needed for a complete professional rotation system

Crossfade / Transition Intelligence

- AutoDJ does basic crossfading but no energy-aware transitions
- Enterprise systems match the outro energy of track A to the intro energy of track B
- Relatively low effort if audio analysis metadata (tempo, energy tags) is already in the library

Weekly Grid View for Wheel Assignments

- PR7 live tab is read-only monitoring of the current hour — good
- No visual weekly grid showing which template covers which daypart block across the full week
- PDs think in weekly grids — Monday 6am-10am Morning Drive, Saturday all-day Weekend — not lists

Template Export / Import as File

- PR10 copying creates an independent copy within a station
- No way to export a template as a portable file and import it into another AzuraCast installation
- Useful for anyone running multiple internet stations with similar formats

5 · Station ID at Top of Hour — Problem and Fix

The Problem

The clock wheel fires a hard anchor at position 0 — top of hour — but the track already playing when the hour turns will finish first. The station ID plays late, sometimes 30 to 90 seconds past the hour depending on what was playing. This is the most noticeable operational issue for any station running structured hourly programming.

The Right Fix — End-of-Hour Lookahead ■

- The scheduler looks ahead before the hour ends
- It selects the last track of the outgoing hour to finish at or before the top of the next hour
- If no track fits cleanly it picks a shorter candidate from the same category
- The ID fires exactly at 00:00

What needs building:

- End-of-hour lookahead pass in `ClockWheelPlaybackPlanner` — currently looks forward from current position only, not backward from the hour boundary
- A fit-to-boundary selection pass specifically for the final music slot of each hour
- Contained change to PR3 planner logic — does not affect anything else in the engine

The Compromise Fix — Fade Cap on Final Slot ■

- For the last slot of the hour only, enforce a fade-out cap so ID fires within a tolerance window (e.g. no more than 15 seconds late)
- `ClockWheelAnnotator` already handles `cue_out` caps — extend that logic to the end-of-hour boundary
- Use this as fallback only if the lookahead approach cannot be implemented first

Legal ID Slot Type ■

- A `legal_id` slot type distinct from the generic `id` type
- Mandatory flag — this slot cannot be skipped or substituted by AutoDJ under any circumstances
- Separate audit logging in `clock_wheel_events` so you can prove the ID fired and exactly when
- Sequential rotation algorithm so multiple ID recordings rotate evenly — random can repeat
- Top of hour ID compliance log: expected fire time vs actual, weekly percentage report

6 · Realistic Advertising for Internet Radio

These are the three realistic paths for internet radio. No FM transmission requirements, no FCC traffic logs. All three can coexist — start with Option 2 (smallest build), add Option 3, then invest in Option 1 when the audience justifies it.

Option 1 — Programmatic Ads via Triton Digital / AdsWizz ■ (Biggest Revenue)

How real internet radio stations run ads at scale. BBC Sounds, iHeart, Audacy all use Triton or AdsWizz. The only path to real CPM revenue.

How it works:

- Ad clock wheel slot fires
- AzuraCast calls a VAST audio endpoint instead of pulling a local file
- Triton/AdsWizz returns a targeted audio ad — listener location, time of day, demographics
- Ad plays, clock wheel resumes normally
- Impression reported back to the ad server — required for getting paid

What needs building:

- Ad slot type that makes an HTTP call to a VAST audio endpoint instead of pulling local media
- Parse VAST response and inject returned audio URL into the queue
- Report impression back to ad server on play
- No-fill fallback — play a house ad or promo if no ad is returned

Option 2 — Sponsor / House Ad Rotation ■ (Easiest to Build)

How small and medium internet stations sell presenting sponsor packages without programmatic complexity. Already 80% possible with what you have.

- Dedicated sponsors playlist per daypart
- Clock wheel ad slot draws from it on rotation
- Burn rate protection already ensures sponsors rotate fairly — PR9 handles this
- Only gap: sponsor management UI — upload spot, assign to daypart, set rotation weight, set flight dates (start and end date the spot is active)

Option 3 — Pre-Roll on Stream Join ■ (Quick Win)

When a new listener tunes in, play a short ad before they hit the live stream. Very common on internet radio. Small build.

- AzuraCast already has intro files per mountpoint
- Gap: make that intro dynamic — pulling from the sponsor pool or VAST endpoint

- Small build on top of existing mountpoint intro functionality

7 · AzuraCast Analytics — Complete Build Plan

Foundation — Listener Session Table ■ (Build This First)

Every audience analytics feature below depends on this table. AzuraCast logs raw connections but does not aggregate them into sessions. Without this you have no audience data to show sponsors or act on.

Column	Description
session_id	Auto-increment primary key
listener_ip_hash	Hashed for privacy — no raw IPs stored
mountpoint	Which stream quality / mount point the listener connected to
connected_at	Timestamp when listener joined
disconnected_at	Timestamp when listener left
duration_seconds	Calculated session length
country / city	Derived from IP at connection time
user_agent	Player or app used to connect
referrer	Where the listener came from — TuneIn, website, direct, app

High Value Analytics ■

Daypart Performance Dashboard

- Average concurrent listeners per daypart across the week
- Best and worst performing dayparts by audience size and retention
- Trend over time — is Morning Drive growing week over week
- Exportable for sponsor proposals

Top of Hour ID Compliance Log

- Did the station ID fire within tolerance of 00:00 this hour
- Expected fire time vs actual fire time logged in clock_wheel_events for every ID slot
- Weekly compliance report — percentage of IDs fired within 15 seconds of top of hour

Clock Performance vs Audience Correlation

- PR12 already logs every fallback, drift, and burn rate warning
- Cross with listener session concurrency — did listeners drop when AutoDJ fallback spiked
- Unique to your platform — no other open source radio tool connects scheduling quality to real listener impact

Medium Value Analytics ■

Hourly Listener Heatmap

- Listener concurrency mapped across every hour of the week
- Shows real peak hours vs what your daypart schedule currently assumes
- Classic radio tool — PDs use this to decide where to put their best content

Song Dropout Tracking

- Listener disconnects within first 30 seconds of a track starting — likely a skip signal
- Aggregate across all plays of each track over 30 days
- Surfaces worst-performing tracks without needing expensive music research
- Over time becomes a feedback loop into rotation weighting — self-improving system

Sponsor Spot Reporting

- Impressions served per sponsor spot
- Which daypart delivered the most plays
- Simple export for billing and proof of performance to sponsors

Geographic Audience Breakdown

- Where listeners are connecting from — IP data already exists in AzuraCast, just not aggregated
- Useful for ad sales — '70% of our audience is US-based'
- Also useful for scheduling — if most listeners are in a different timezone to the server

Mount Point Breakdown

- Which stream quality listeners actually use — 128k vs 64k vs mobile mount
- Informs infrastructure and bandwidth decisions
- Shows whether investing in a higher quality mount is justified by actual usage

Lower Priority Analytics ■

Referrer Tracking

- Which platform sent the listener — TuneIn, your website, a mobile app, direct URL
- Tells you where to focus promotional effort

New vs Returning Listener Ratio

- By hashed IP — are you growing a new audience or just retaining existing listeners
- Week over week trend shows whether outreach is working

Week over Week Listener Growth

- Total unique sessions per week charted over time
- The single number that tells you whether the station is growing

Average Session Duration Trend

- Are listeners staying longer over time — measures content quality improvement
- Breakdown by daypart shows which programming keeps people tuned in longest

8 · Master Build Order — Everything Prioritised

Complete prioritised list of every item discussed. Do items 1–5 and the clock wheel is professional grade. Do through 10 and analytics are better than most paid internet radio platforms. Items 11 onward are meaningful improvements but not blockers.

#	Item	Priority	What to Build	Why
1	PR6 Unit Tests	HIGH	Create tests/Unit, write all test classes	Protects everything already built from regressions
2	Format Clock Category Enforcement	HIGH	Hard constraint on slot playlist targeting	Makes clock wheels actually programmable by a PD
3	End-of-Hour Lookahead	HIGH	Planner backward pass from hour boundary	Fixes station ID running late — core correctness
4	Legal ID Slot Type	HIGH	Mandatory type, sequential rotation, compliance log	Station ID as first-class scheduling concern
5	Listener Session Table	HIGH	New session DB table + connection hooks	Foundation for all audience analytics — everything below needs this
6	Per-Slot Separation Overrides	MED	Separation config at slot level not just wheel	Completes separation rules to true professional level
7	Rotation Goals (Positive Scheduling)	MED	Positive scheduling targets per category per day	Completes rotation — burn rate is ceiling, goals are floor
8	Daypart Audience Report	MED	Concurrency per daypart, trend, exportable	First thing a sponsor asks for
9	Sponsor Management UI	MED	Upload spot, flight dates, rotation weight per daypart	Immediate revenue path — small build on existing PR9
10	Clock Performance vs Audience	MED	PR12 data crossed with session concurrency	Unique differentiator — no other OSS platform has this
11	Hard Anchor Miss Recovery	MED	Active response + logging on missed hard anchors	Operational resilience
12	Top of Hour ID Compliance Log	MED	Expected vs actual fire time, weekly % report	Proves ID scheduling accuracy over time

1 3	Hourly Listener Heatmap	MED	Concurrency grid across every hour of the week	Validate daypart assumptions against real audience
1 4	Song Dropout Tracking	MED	Disconnects in first 30s per track, 30-day aggregate	Self-improving rotation input over time
1 5	Crossfade / Transition Intelligence	MED	Energy-aware track transitions using audio metadata	Improves listen-through — requires energy tags in library
1 6	Sponsor Spot Reporting	LOW	Impressions per spot, daypart delivery, export	Essential once paying sponsors are on board
1 7	Geographic Audience Breakdown	LOW	Aggregate IP data by country/city	Ad sales and scheduling timezone decisions
1 8	Mount Point Breakdown	LOW	Usage per stream quality / mount	Infrastructure and bandwidth planning
1 9	VAST / Triton Ad Integration	LOW	Ad slot calls VAST endpoint, reports impression, no-fill fallback	Real CPM revenue — bigger build, needs audience first
2 0	Weekly Grid View	LOW	Visual daypart / template assignment grid for full week	PD workflow quality of life
2 1	Template Export / Import as File	LOW	Portable template file format, import to any install	Multi-station operators running similar formats
2 2	Referrer Tracking	LOW	Source attribution per listener session	Focus promotional effort on what works
2 3	New vs Returning Listener Ratio	LOW	By hashed IP, week over week	Measures whether outreach is growing real audience
2 4	Pre-Roll on Stream Join	LOW	Dynamic intro from sponsor pool or VAST on connect	Common on internet radio — small build on existing mountpoint intro